Examining the Effects of Sex and Education on Tau Pathology and Neuronal Preservation in Alzheimer's Disease

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BACKGROUND

Analysis 1

Grouping Variable 1: Sex (Two bins: Male & Female). In a 2020 study by Sundermann et al., sex differences in Alzheimer's disease pathology were documented, with research showing that females exhibit higher levels of tau biomarkers than males, potentially influenced by hormonal factors such as testosterone (Sundermann, E. E., Panizzon, M. S., Chen, X., Andrews, M., Galasko, D., Banks, S. J., & Alzheimer's Disease Neuroimaging Initiative (2020). Sex differences in Alzheimer's-related Tau biomarkers and a mediating effect of testosterone. *Biology of sex differences*, *11*(1), 33).

Dependent Variable 1: Percent AT8 positive area_Grey matter, also known as tau pathology, refers to the abnormal accumulation, misfolding, and aggregation of tau protein in the brain, forming neurofibrillary tangles that disrupt neuronal function and cause cell death. It acts as a specific marker for Alzheimer's disease pathology by detecting abnormal, hyperphosphorylated tau protein and is primarily used in research and diagnostics to visualize neurofibrillary tangles (NFTs) and "pre-tangles" in brain tissue sections.

Dependent Variable 2: Percent NeuN positive area_Grey matter measures neuronal preservation, essentially how much tissue is neuron positive and how many neurons are still there). It is a marker that visualizes neuron density, detecting its reduction to show neuronal loss in diseased tissues (e.g. neurodegeneration).

Hypothesis: Female individuals will exhibit greater neurodegenerative pathology compared to Male individuals. Specifically, Female individuals will show higher percent AT8 positive area (indicating increased tau pathology) and lower percent NeuN positive area (indicating reduced neuronal preservation) in grey matter.

Analysis 2

Grouping Variable 1: Years of Education (Two bins: Low & High). Educational attainment is widely used as a proxy for cognitive reserve, a concept that explains individual differences in susceptibility to age-related brain changes and Alzheimer's disease pathology. Cognitive reserve suggests that some individuals can tolerate greater levels of neuropathology while maintaining cognitive function. One study by Yaakov Stern showed that lifelong experiences – particularly educational and occupational attainment – contribute to this reserve, and that individuals with higher levels of education have a reduced risk of developing Alzheimer's disease and may demonstrate greater resilience to neurodegenerative processes (Stern Y (2012). Cognitive reserve in ageing and Alzheimer's disease. *The Lancet. Neurology*, *11*(11)). In a large prospective cohort study using UK Biobank data, it was also found that higher educational attainment is associated with a significantly reduced risk of Alzheimer's disease, with individuals in the high education group showing about a 30% lower risk compared to those with low education. Importantly, this protective effect persisted regardless of genetic risk, suggesting that education may independently contribute to resilience against disease development. These findings support the role of education as a key factor in cognitive reserve and justify its use as a grouping variable when examining differences in neurodegenerative pathology (Li, X., Zhang, Y., Zhang, C., Zheng, Y., Liu, R., & Xiao, S. (2023). Education counteracts the genetic risk of Alzheimer's disease without an interaction effect. *Frontiers in public health*, *11*, 1178017).

Dependent Variable 1: Percent AT8 positive area_Grey matter.

Dependent Variable 2: Percent NeuN positive area_Grey matter.

Hypothesis: Individuals with higher levels of education will exhibit reduced neurodegenerative pathology compared to those with lower levels of education. Specifically, individuals in the high education group are expected to show lower percent AT8 positive area (indicating reduced tau pathology) and higher percent NeuN positive area (indicating greater neuronal preservation) in grey matter, consistent with the cognitive reserve hypothesis and the protective effects of education on brain resilience.

METHODS

```python
# Import relevant modules

import pandas as pd
import matplotlib.pyplot as plt
from scipy import stats

# Load dataset

alzheimers_IDs = pd.read_csv('https://raw.githubusercontent.com/scottac1111/Alex-Scott-Repository/refs/heads/main/alzheimers_IDs_final_project%20(4).csv')

alzheimers_pathology = pd.read_csv('https://raw.githubusercontent.com/scottac1111/Alex-Scott-Repository/refs/heads/main/alzheimers_pathology_final_project%20(2).csv')
```

```python
# Extracted only the IVs and DVs I will be using

alzheimers_relevant_data = pd.read_csv('https://raw.githubusercontent.com/scottac1111/Alex-Scott-Repository/refs/heads/main/Alzheimers%20data%20(5%3A4).csv')
df = pd.DataFrame(alzheimers_relevant_data)
```

```python
# Define variables

sex = df['Sex']
education = df['Years of education']

at8 = df['percent AT8 positive area_Grey matter']
neun = df['percent NeuN positive area_Grey matter']
```

```python
# Bargraphs showing the distribution of IVs

# Sex distribution
sex_counts = sex.value_counts()

plt.bar(sex_counts.index, sex_counts.values)
plt.xlabel('Sex')
plt.ylabel('Number of participants')
plt.title('Distribution of Sex')
plt.show()


# Years of Education distribution

median_edu = education.median() #median is 16

df['Education_Group'] = df['Years of education'].apply(
    lambda x: 'Low' if x < median_edu else 'High'
)

edu_counts = df['Education_Group'].value_counts()

plt.bar(edu_counts.index, edu_counts.values)
plt.xlabel('Years of Education')
plt.ylabel('Number of participants')
plt.title('Distribution of Education Group')
plt.show()
```

```python
# Boxplots for NeuN (Male vs Female)

male_NeuN = df[df['Sex'] == 'Male']['percent NeuN positive area_Grey matter'].dropna()
female_NeuN = df[df['Sex'] == 'Female']['percent NeuN positive area_Grey matter'].dropna()

plt.boxplot([male_NeuN, female_NeuN])
plt.xticks([1, 2], ['Male', 'Female'])

plt.ylabel('Percent NeuN positive area_Grey matter')
plt.title('NeuN by Sex')

# Use a p-test if skewness is normal and ranksum test if skewness is not normal
p_skew = stats.skewtest(neun.dropna())[1]

if p_skew < 0.05:
    stat_NeuN, p_value_NeuN = stats.ranksums(male_NeuN, female_NeuN)
    print('NeuN (rank-sum):', stat_NeuN, p_value_NeuN)
else:
    stat_NeuN, p_value_NeuN = stats.ttest_ind(male_NeuN, female_NeuN)
    print('NeuN (t-test):', stat_NeuN, p_value_NeuN)

plt.show()
```

```python
# Boxplots for NeuN (Years of Education: High vs Low)

low_NeuN = df[df['Education_Group'] == 'Low']['percent NeuN positive area_Grey matter'].dropna()
high_NeuN = df[df['Education_Group'] == 'High']['percent NeuN positive area_Grey matter'].dropna()

plt.boxplot([low_NeuN, high_NeuN])
plt.xticks([1, 2], ['Low', 'High'])

plt.ylabel('Percent NeuN positive area_Grey matter')
plt.title('NeuN by Education Level')

# Use a p-test if skewness is normal and ranksum test if skewness is not normal

p_skew = stats.skewtest(neun.dropna())[1]

if p_skew < 0.05:
    stat_NeuN, p_value_NeuN = stats.ranksums(low_NeuN, high_NeuN)
    print('NeuN (rank-sum):', stat_NeuN, p_value_NeuN)
else:
    stat_NeuN, p_value_NeuN = stats.ttest_ind(low_NeuN, high_NeuN)
    print('NeuN (t-test):', stat_NeuN, p_value_NeuN)

plt.show()
```

```python
# Using a loop to plot each datapoint according to Years of Education

# Clean data for plotting

clean_df = df[['Education_Group',
               'percent AT8 positive area_Grey matter',
               'percent NeuN positive area_Grey matter']].dropna()

edu_list = np.array(clean_df['Education_Group'])
at8_clean = np.array(clean_df['percent AT8 positive area_Grey matter'])
neun_clean = np.array(clean_df['percent NeuN positive area_Grey matter'])

for i in range(len(edu_list)):

    if edu_list[i] == 'High':
        plt.scatter(at8_clean[i], neun_clean[i], color='blue')

    elif edu_list[i] == 'Low':
        plt.scatter(at8_clean[i], neun_clean[i], color='red')

plt.xlabel('Percent AT8 positive area_Grey matter')
plt.ylabel('Percent NeuN positive area_Grey matter')
plt.title('AT8 vs NeuN by Education Level')
plt.show()
```

```python
# Histogram of 1st dependent variable (percent AT8 positive area_Grey matter)

plt.hist(at8)
plt.xlabel('Percent AT8 positive area_Grey matter')
plt.ylabel('Number of subjects')
plt.title('AT8 Distribution')
plt.show()

stat, p = stats.skewtest(at8)
print('AT8 skewness p =', p)
```

```python
# Histogram of 2nd dependent variable (percent NeuN positive area_Grey matter)

plt.hist(neun)
plt.xlabel('Percent NeuN positive area_Grey matter')
plt.ylabel('Number of subjects')
plt.title('NeuN Distribution')
plt.show()

stat, p = stats.skewtest(neun)
print('NeuN skewness p =', p)
```

```python
# Boxplot for AT8 (Male vs Female)

male_AT8 = df[df['Sex'] == 'Male']['percent AT8 positive area_Grey matter'].dropna()
female_AT8 = df[df['Sex'] == 'Female']['percent AT8 positive area_Grey matter'].dropna()

plt.boxplot([male_AT8, female_AT8])
plt.xticks([1, 2], ['Male', 'Female'])

plt.ylabel('Percent AT8 positive area_Grey matter')
plt.title('AT8 by Sex')

# Use a p-test if skewness is normal and ranksum test if skewness is not normal
p_skew = stats.skewtest(at8.dropna())[1]

if p_skew < 0.05:
    stat_AT8, p_value_AT8 = stats.ranksums(male_AT8, female_AT8)
    print('AT8 (rank-sum):', stat_AT8, p_value_AT8)
else:
    stat_AT8, p_value_AT8 = stats.ttest_ind(male_AT8, female_AT8)
    print('AT8 (t-test):', stat_AT8, p_value_AT8)

plt.show()
```

```python
# Boxplot for AT8 (Years of Education: high vs Low)


low_AT8 = df[df['Education_Group'] == 'Low']['percent AT8 positive area_Grey matter'].dropna()
high_AT8 = df[df['Education_Group'] == 'High']['percent AT8 positive area_Grey matter'].dropna()

plt.boxplot([low_AT8, high_AT8])
plt.xticks([1, 2], ['Low', 'High'])

plt.ylabel('Percent AT8 positive area_Grey matter')
plt.title('AT8 by Education Level')

# Use a p-test if skewness is normal and ranksum test if skewness is not normal

p_skew = stats.skewtest(at8.dropna())[1]

if p_skew < 0.05:
    stat_AT8, p_value_AT8 = stats.ranksums(low_AT8, high_AT8)
    print('AT8 (rank-sum):', stat_AT8, p_value_AT8)
else:
    stat_AT8, p_value_AT8 = stats.ttest_ind(low_AT8, high_AT8)
    print('AT8 (t-test):', stat_AT8, p_value_AT8)

plt.show()
```

```python
import numpy as np

# Using a loop to plot each datapoint according to sex
# Also storing values in separate arrays

sex_list = np.array(df['Sex'])

male_AT8 = []
male_NeuN = []
female_AT8 = []
female_NeuN = []

for i in range(len(sex_list)):

    if sex_list[i] == "Male":
        plt.scatter(at8[i], neun[i], color="blue")
        male_AT8.append(at8[i])
        male_NeuN.append(neun[i])

    elif sex_list[i] == "Female":
        plt.scatter(at8[i], neun[i], color="red")
        female_AT8.append(at8[i])
        female_NeuN.append(neun[i])


plt.xlabel('Percent AT8 positive area_Grey matter')
plt.ylabel('Percent NeuN positive area_Grey matter')
plt.title('AT8 vs NeuN by Sex')
plt.show()
```

RESULTS

Analysis 1

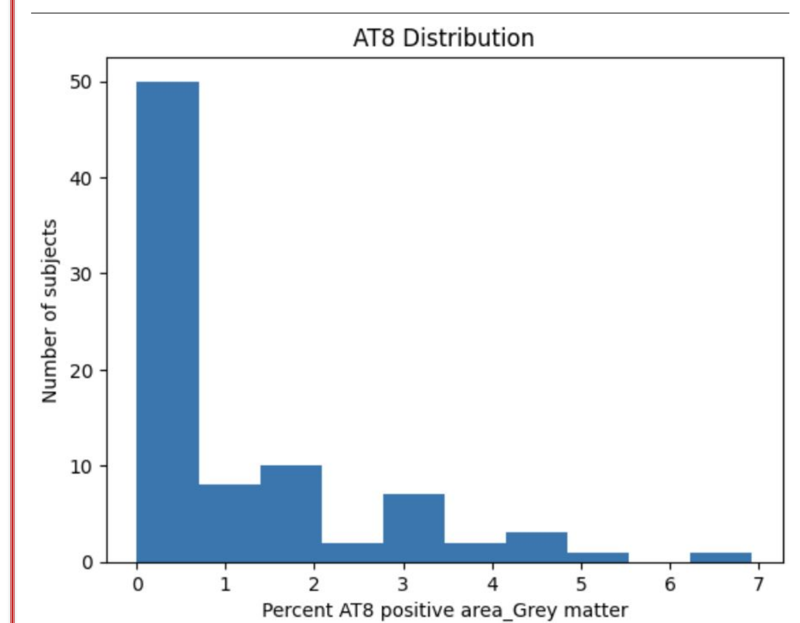

Fig. 1 The distribution of percent AT8 positive area in grey matter is rightly skewed, with most subjects clustered at low values and a smaller number of subjects with higher tau pathology. A skewness test confirmed that the data were not normally distributed (p = 7.65×10^{-7}).

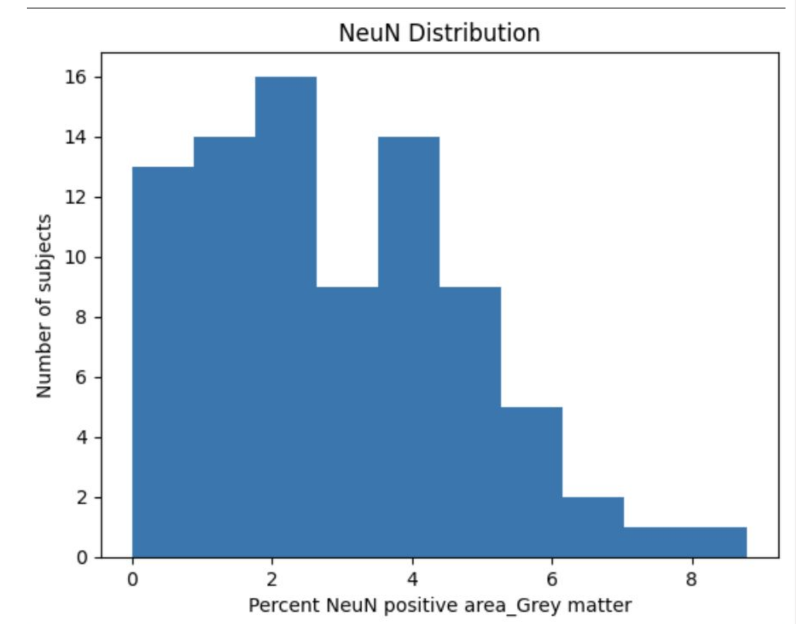

Fig. 2 The distribution of percent NeuN positive area in grey matter is moderately left-skewed with a broad spread of values across subjects. A skewness test indicated that the data deviated from normality (p = 0.038), suggesting a non-normal distribution.

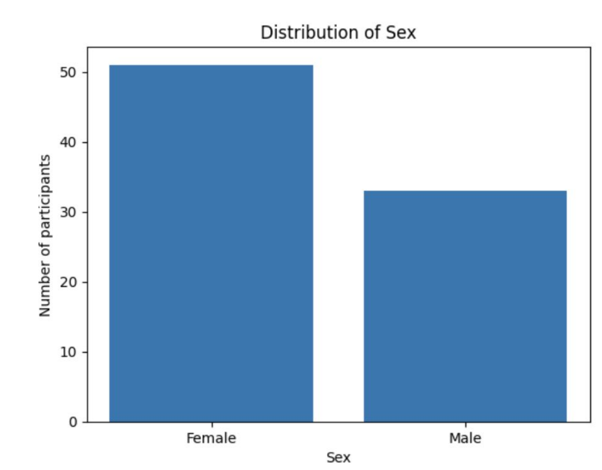

Fig. 3 Distribution of Sex (IV). Male participants (n=33), Female participants (n=51).

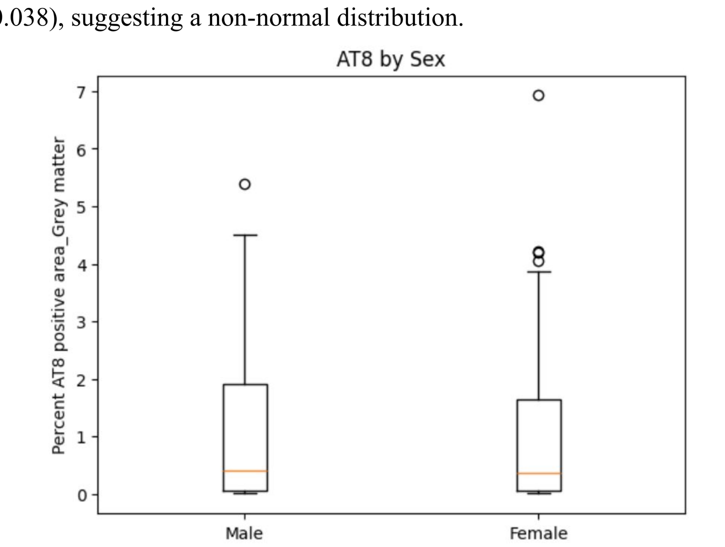

Fig. 4 Wilcoxon Rank-Sum Statistic: 0.041. P-value > 0.05 (p = 0.967). There is not enough evidence to reject the null hypothesis, indicating no significant difference in percent AT8 positive area between male and female individuals.

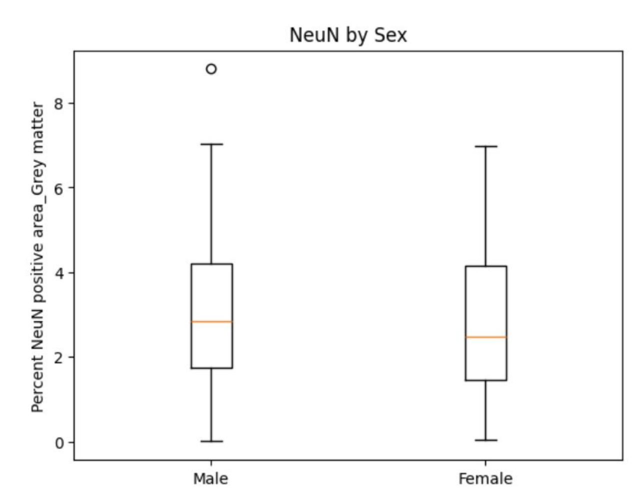

Fig. 5 Wilcoxon Rank-Sum Statistic: 0.453. P-value > 0.05 (p = 0.650). There is not enough evidence to reject the null hypothesis, indicating no significant difference in percent NeuN positive area between male and female individuals.

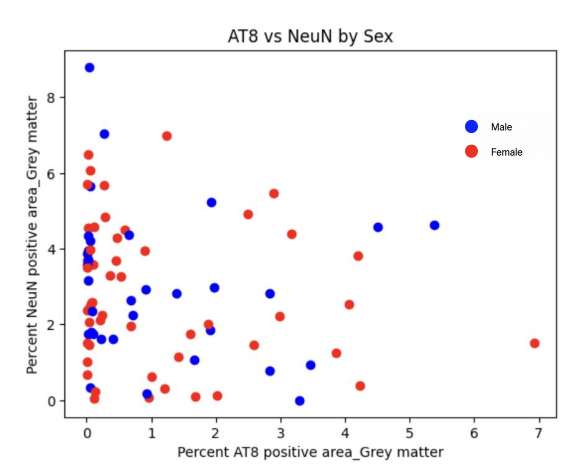

Fig. 6 The data appears to be mostly clustered in one area, and while the case could be made that there is a somewhat moderately positive association, there doesn't seem to be a clear linear relationship between AT8 and NeuN, with the distribution of points suggesting a weak or absent correlation between tau pathology and neuronal preservation. Male and female subjects showed substantial overlap across the range of values, indicating no strong sex-based differences in this relationship

Analysis 2

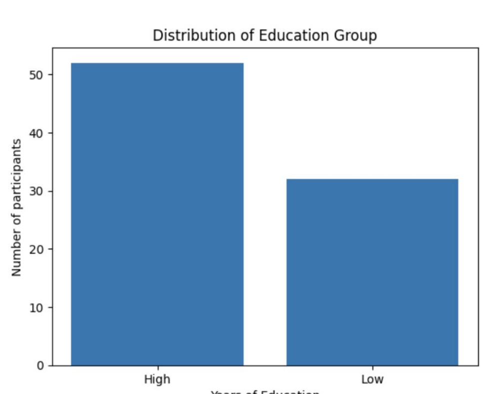

Fig. 7 Distribution of Years of Education (IV). Number of participants with less than 16 years of education (median) is n=32. Number of participants who received 16 years or greater of education is n=52.

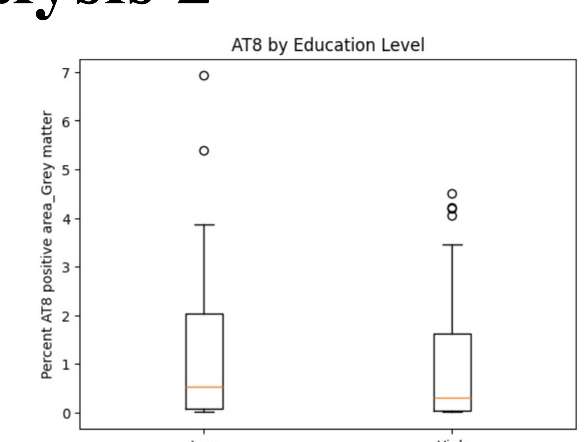

Fig. 8 Wilcoxon Rank-Sum Statistic: 0.986. P-value > 0.05 (p = 0.324). There is not enough evidence to reject the null hypothesis, indicating no significant difference in percent AT8 positive area between low and high education groups.

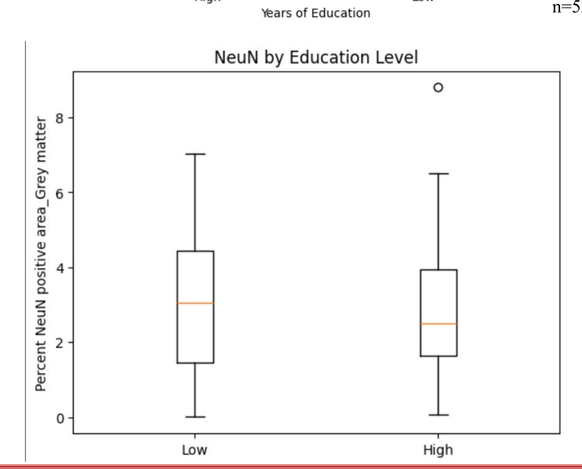

Fig. 9 Wilcoxon Rank-Sum Statistic: 0.433. P-value > 0.05 (p = 0.665). There is not enough evidence to reject the null hypothesis, indicating no significant difference in percent NeuN positive area between low and high education groups.

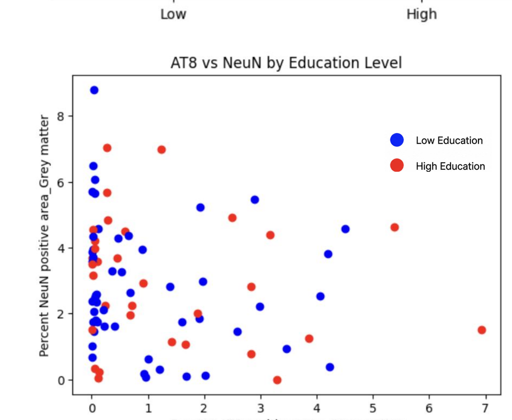

Fig. 10 The data showed no strong linear relationship between AT8 and NeuN, and both education groups overlapped considerably, suggesting that education level does not strongly alter the relationship between tau pathology and neuronal preservation.

CONCLUSIONS

Analysis 1: Sex Differences in Tau Pathology and Neuronal Preservation

Overall, my results did not support the original hypothesis that females would exhibit greater neurodegenerative pathology than males. While prior literature suggests sex-related differences in tau accumulation, my data found no statistically significant differences between male and female participants in either percent AT8 positive area (tau pathology) or percent NeuN positive area (neuronal preservation). Additionally, the relationship between AT8 and NeuN appeared weak or absent, with no clear linear correlation and substantial overlap between sexes. This suggests that, within this dataset, tau pathology and neuronal density may not be directly coupled, or that their relationship is more complex than a simple inverse association.

These findings are important because they highlight that sex alone may not be a strong determinant of tau burden or neuronal loss in this sample. This could reflect sample size limitations, variability in disease progression, or the influence of other unmeasured factors (e.g., age, genetics, or hormonal status). More broadly, it emphasizes the need to consider multifactorial influences when studying Alzheimer's disease pathology rather than relying on single demographic variables.

Analysis 2: Education Level and Neurodegenerative Pathology

My results also did not support the hypothesis that higher education would be associated with reduced tau pathology and greater neuronal preservation. There were no significant differences between low and high education groups for either AT8 or NeuN measures. Furthermore, the scatterplot analysis showed no strong relationship between tau pathology and neuronal preservation, and both education groups overlapped substantially. This suggests that education level did not meaningfully alter the relationship between pathology and neuronal density in this dataset.

These findings are notable because they contrast with the well-established theory of cognitive reserve, which proposes that higher education confers resilience against neurodegeneration. One possible interpretation is that cognitive reserve may not reduce the presence of pathology itself, but rather helps individuals maintain cognitive function despite pathology.

Overall, this analysis suggests that while education is an important factor in Alzheimer's disease outcomes, its effects may be functional rather than structural, influencing cognition more than measurable pathology.

FUTURE DIRECTIONS

While my analyses did not identify significant effects of sex or education on tau pathology (AT8) or neuronal preservation (NeuN), several additional approaches could help clarify these relationships. First, future studies should use larger and more diverse sample sizes, since having a smaller sample may have made it harder to detect smaller differences between groups.

Including additional demographic and biological variables – such as age, genetic risk factors (e.g., APOE genotype), and disease stage – would also allow for a more comprehensive understanding of how multiple factors interact to influence Alzheimer's pathology.

Second, further analyses could examine regional differences in the brain, rather than focusing only on overall grey matter measures. Tau accumulation and neuronal loss are known to occur unevenly across brain regions, so region-specific analyses may reveal patterns that are not detectable at a global level.

Third, because my study focused on structural and pathological markers, future work should integrate cognitive or behavioral data. This would help test the idea that factors like education contribute to cognitive reserve, allowing individuals to maintain function despite similar levels of pathology. Longitudinal studies would be especially valuable in this context, as they could track how pathology and neuronal loss evolve over time and whether education or sex influences the rate of decline.

Finally, more advanced statistical approaches – such as regression models or interaction analyses – could be used to examine whether variables like education moderate the relationship between tau pathology and neuronal preservation. Together, these approaches would provide a more nuanced understanding of Alzheimer's disease and help identify factors that may protect against its progression.